

Gus's Workboard: Grammar Contraption Parts

A list of silly, physics-style, plug-and-play machine parts for Gus's Workboard. Each part does something funny on the board and has a real grammar job hidden inside it. Prepared by Claudia (game and instructional design review), 2026-10-07.

Teacher's request: "claudia, lets develop a list of features that can be physics/contraption/rube goldberg fun plug and play machine elements that are rooted in grammatical concepts"

Teacher's direction for the whole Workboard: "i want this to be a sandbox style for the kids to play around with grammar in the same STEAM style way they like to play other games, with hidden grammar concepts"

Build tiers:

- **NOW:** fits the current grammar engine.
- **SOON:** needs a small engine addition.
- **LATER:** needs grammar the engine does not teach yet.

Rules every part follows

- **Wrong is private.** A wrong placement makes the part wobble, clunk or jam on the student's own screen only. Nobody else sees it.
- **Effort pays.** A fixed jam still earns the full run and the movie.
- **No gambling.** There are no spinning prize wheels, loot boxes or random rewards. The grammar causes every reaction.
- **iPad first.** Every part snaps on by tap or drag. Targets are 44px or larger, and nothing needs hover.
- **Set per student.** The teacher can turn parts on or off.
- **Silly is welcome.** Sounds, bounces, sparks and steam are all good.

Already on the Workboard (shipped 2026-10-07)

Every fun part now has a grammar job:

Part	What it does	Hidden grammar
Big Horn (new)	HONK! Blasts at the end	Exclamation point, or the "!" right after an interjection
Spring Mat	BOING! Bounces in a word	Adjective (describing word)
Fan	WHOOSH! Blows in a word	Adverb (how word)
Ramp	WHEEE! Rolls in a word	Preposition (where word)
Conveyor Belt	CLANK-CLANK, carries a word	Conjunction (joining word)
Bucket Drop	SPLOSH! Drops in a word	Article (a, an, the)

Part	What it does	Hidden grammar
Pulley	HEAVE-HO! Hoists the next word	Capital letter
Bell	DING!	Period
Dominoes	CLACK-CLACK	Comma
Duplicator (new)	COPY! COPY! Copies the next noun	Plural noun, and the verb has to agree

Shipped 2026-10-07: every NOW part

All 14 NOW parts below are on the Workboard, plus a rule from the teacher: parts only run when they are snapped in the right order and the sentence is correct. The Clock now changes the action words to past, present or future.

1. Capital letters and punctuation

1. Proper Name Stamp (SOON)

- **What it does:** A giant rubber stamp hangs over the pipe. When a name passes, it slams down and a gold capital letter pops up with a clang.
- **Grammar job:** Names of people, places and days start with a capital letter.
- **Trains:** Proper nouns.
- **Example:** "gus visited vermont" becomes "Gus visited Vermont."

2. Mood Meter Valve (NOW, shipped)

- **What it does:** A pressure gauge whose needle swings from "calm" to "BIG FEELING." It opens the Bell or the Big Horn.
- **Grammar job:** Matches the end punctuation to the sentence. Calm gets a period and a strong feeling gets an exclamation point.
- **Trains:** Choosing end punctuation on purpose.
- **Example:** "The roof is on fire." becomes "The roof is on fire!"

3. Comma List Train (SOON)

- **What it does:** A little train with one cart per item. Each coupler between carts is a comma, and the last cart hooks on with an "and" hook.
- **Grammar job:** Commas between items in a list.
- **Trains:** Commas in a series.
- **Example:** "I packed pizza chips and cake" becomes "I packed pizza, chips, and cake."

4. Sentence Sprinter Flag (NOW, shipped)

- **What it does:** A starting gun fires at the first word and a checkered flag drops at the last mark.

- **Grammar job:** A capital letter at the start and punctuation at the end.
- **Trains:** Sentence boundaries.
- **Example:** "the dog barked" gets a Pulley and a Bell, then the flag waves.

2. Nouns and articles

5. Irregular Trapdoor (SOON)

- **What it does:** The Duplicator's weird cousin. The marble drops through a trapdoor and comes out odd: a cartoon "mouses" gets crushed and "mice" comes out.
- **Grammar job:** Irregular plurals.
- **Trains:** Irregular plural nouns.
- **Example:** "two mouse" becomes "two mice."

6. A/An Vowel Sniffer (NOW, shipped)

- **What it does:** A vacuum nozzle on a spring sniffs the next word's first letter. A vowel sound makes it sneeze and swap the Bucket Drop's "a" for "an."
- **Grammar job:** a versus an.
- **Trains:** Articles before vowel sounds.
- **Example:** "a apple" becomes "an apple."

3. Pronouns

7. Pronoun Teleporter (NOW, shipped)

- **What it does:** A glowing yellow pad. When a noun repeats, the second one steps on it and zaps into a pronoun.
- **Grammar job:** Replaces a repeated noun with the right pronoun.
- **Trains:** Pronoun reference.
- **Example:** "Mia fed the cat. Mia smiled." becomes "Mia fed the cat. She smiled."

8. Subject and Object Turnstile (SOON)

- **What it does:** A two-way gate. The doer side lets I, he and she through. The receiver side lets me, him and her through. A wrong pronoun makes the gate spin and bonk.
- **Grammar job:** Pronoun case.
- **Trains:** Subject pronouns and object pronouns.
- **Example:** "Him threw the ball to I" becomes "He threw the ball to me."

9. Possessive Tag Gun (LATER)

- **What it does:** A sticker gun shoots a name tag onto an item.

- **Grammar job:** An apostrophe and "s" show who owns something.
- **Trains:** Possessives.
- **Example:** "the dogs bone" becomes "the dog's bone."

4. Verbs and time

10. Agreement Gears (NOW, shipped)

- **What it does:** The noun machine and the action machine each get a gear. Matching gears mesh with a smooth whir. A mismatch grinds and throws sparks.
- **Grammar job:** Subject and verb agreement.
- **Trains:** One and more than one.
- **Example:** "The dogs runs" becomes "The dogs run."

11. Time Tunnel (NOW, shipped)

- **What it does:** A three-lane tunnel with a past, present and future gate. The action machine changes shape as the marble passes through.
- **Grammar job:** The verb's time matches the Clock and any time word.
- **Trains:** Consistent verb tense.
- **Example:** "Last night, we walk" becomes "Last night, we walked."

12. Irregular Past Press (SOON)

- **What it does:** A heavy press. "goed" goes in and "went" gets stamped out.
- **Grammar job:** True irregular past forms.
- **Trains:** Irregular past verbs.
- **Example:** "He runned home" becomes "He ran home."

5. Adjectives and adverbs

13. Describe Sorter Chute (NOW, shipped)

- **What it does:** A chute with slots in order (opinion, size, color). The words roll in and clack into their slots.
- **Grammar job:** Describing words in the right order.
- **Trains:** Adjective order.
- **Example:** "a red big ugly truck" becomes "an ugly big red truck."

14. How-Dial (NOW, shipped)

- **What it does:** A big dial on the Fan. "slowly" makes the marble creep and "quickly" makes it zoom.

- **Grammar job:** How the action happens, with the adverb beside its verb.
- **Trains:** Adverbs of manner.
- **Example:** "The turtle moved slowly."

15. Opener Slingshot (NOW, shipped)

- **What it does:** A slingshot flings a how word to the front of the line, and a tiny comma domino tips behind it.
- **Grammar job:** An adverb opener with its comma.
- **Trains:** Sentence openers.
- **Example:** "Suddenly, the lights went out."

16. Size-Up Inflator (LATER)

- **What it does:** A bike pump. Each pump grows the describing word's balloon: tall, taller, tallest.
- **Grammar job:** Comparing forms.
- **Trains:** Comparatives and superlatives.
- **Example:** "Gus is the tall robot" becomes "Gus is the tallest robot."

6. Prepositions

17. Where-To Switchyard (NOW, shipped)

- **What it does:** A rail switch with three paths. The marble goes over a bridge, under a tunnel or through a ring, depending on the preposition.
- **Grammar job:** The preposition shows where.
- **Trains:** Position words.
- **Example:** "The cat jumped over the fence."

18. Dead-End Detector (NOW, shipped)

- **What it does:** A Ramp with no landing gets a "ROAD CLOSED" sign. Add a noun and the landing pad pops into place.
- **Grammar job:** Every preposition needs a noun after it.
- **Trains:** Complete prepositional phrases.
- **Example:** "The cat sat on." becomes "The cat sat on the mat."

7. Conjunctions

19. Merge Funnel (NOW, shipped)

- **What it does:** A big purple funnel. Two noun marbles pour in and one fat marble comes out.

- **Grammar job:** Two nouns joined by "and" count as more than one.
- **Trains:** Compound subjects.
- **Example:** "Tom and Mia run."

20. Comma Hinge Drawbridge (NOW, shipped)

- **What it does:** A drawbridge joins two sentence islands. It only lowers when a comma domino sits at the hinge, right before the joining word.
- **Grammar job:** The comma before and, but or so when two sentences join.
- **Trains:** Compound sentences.
- **Example:** "I was tired, but I kept going."

21. Because Seesaw (LATER)

- **What it does:** A seesaw with a cause on one end and an effect on the other. A "because" weight tips it.
- **Grammar job:** Linking a reason to a result.
- **Trains:** Subordinating conjunctions.
- **Example:** "We stayed in because it rained."

8. Interjections and sentence types

22. Confetti Trapdoor (NOW, shipped)

- **What it does:** An orange popper at the start of the pipe. "Wow!" pops up and confetti falls on the first machine.
- **Grammar job:** An interjection goes first, followed by punctuation.
- **Trains:** Interjections.
- **Example:** "Wow! The marble flew."

23. Question Crane (LATER)

- **What it does:** A hook picks up the action machine and swings it in front of the noun.
- **Grammar job:** Flips the word order and adds a question mark.
- **Trains:** Asking questions.
- **Example:** "You are ready." becomes "Are you ready?"

24. Speech Bubble Blower (LATER)

- **What it does:** A bubble pipe. The words inside get quotation marks, and a tag like "she said" follows.
- **Grammar job:** Quotation marks around spoken words.
- **Trains:** Dialogue punctuation.
- **Example:** "I'm done," she said.

Chain reactions (combos)

- **Plural Pileup.** The Duplicator copies the noun, then the Agreement Gears grind until the verb changes to match: "one dog barks" becomes "two dogs bark."
- **Time Warp.** The Clock set to the future, the Time Tunnel and a time word like "Next" all have to agree. When they do, the movie plays.
- **Dramatic Entrance.** The Confetti Trapdoor, the Mood Meter Valve and the Big Horn: "Wow" opens the line, the gauge reads high, and the Horn blasts the exclamation point.
- **Big Merge.** The Merge Funnel, the Agreement Gears and the Comma Hinge Drawbridge: two nouns pour in, the verb gear shifts to plural, and the bridge lowers to join a second sentence.
- **Road Closed Repair.** The Dead-End Detector and the Where-To Switchyard: add the missing noun, the landing appears, and the marble goes over, under or through on the TV.

Suggested build order

1. **First NOW batch:** Agreement Gears, Time Tunnel, Merge Funnel and the A/An Vowel Sniffer. These pay off the most and each shows grammar through motion.
2. **Second NOW batch:** the rest of the NOW parts.
3. **SOON batch:** Proper Name Stamp, Comma List Train, the Subject and Object Turnstile, and the two irregular parts.
4. **LATER:** after the engine teaches questions, possessives and quotation marks.